Harsh Srivastava

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Education

JSS Academy of Technical Education — Bengaluru, Karnataka
B.E. in Computer Science and Engineering | CGPA: 8.02/10 — *2023 - 2027*

Technical Skills

LLM & Agentic AI: LangChain, LangGraph, Model Context Protocol (MCP), multi-agent orchestration, supervisor workflows, tool calling, guardrails, human-in-the-loop (HITL), workflow state management, prompt engineering

RAG & Retrieval: Retrieval-Augmented Generation (RAG), RAG evaluation, embeddings, semantic search, Qdrant, FAISS, document chunking, top-K retrieval, citation-aware generation

Models & AI Tooling: Hugging Face, Sentence Transformers, BGE embeddings, Tavily Search API, PyPDF ingestion, Trafilatura, BeautifulSoup4, web content extraction

AI Engineering: Python, FastAPI, Streamlit, Pydantic, PostgreSQL, REST APIs, checkpointing, evaluation and review workflows

Deployment & Tools: Docker, Docker Compose, AWS, Git, GitHub

Projects

Offline-Vector RAG Application — GitHub
FastAPI, Qdrant, BGE, Groq, Docker

- Built an end-to-end **RAG pipeline** with recursive document chunking, local **BAAI/bge-small-en-v1.5** embeddings, persistent **Qdrant** vector storage, top-K retrieval, and grounded LLM generation with source citations.
- Developed **FastAPI** health, ingestion, and query endpoints with streaming responses, configurable chunk/retrieval parameters, and idempotent re-ingestion using file-hash, page, and chunk identifiers.
- Containerized the API, Qdrant, and UI with **Docker Compose**; added automated quality checks and documented deployment paths for **AWS EC2, ECS Fargate, ALB, EFS, and GitHub OIDC**.

LangChain Multi-Agent Research System — GitHub
LangChain, Groq, Tavily, Streamlit, Web Extraction

- Engineered a four-stage **multi-agent research pipeline** using two **AgentExecutor** agents for search and reading, followed by dedicated writer and critic chains for structured synthesis and review.
- Built tool-driven web research with **Tavily** search and a resilient extraction pipeline using **Trafilatura**, with **Readability-lxml** and **BeautifulSoup4** fallbacks; persisted intermediate outputs through explicit workflow state.
- Implemented an independent **critic chain** that scores reports, identifies strengths, and suggests improvements; exposed search results, source notes, reports, and quality review through a **Streamlit** workspace.

TripMate AI - Multi-Agent Travel Planner — GitHub
LangGraph, MCP, Guardrails, FastAPI, PostgreSQL, Groq, Docker

- Developed a **LangGraph supervisor workflow** that dynamically routes requests to flight, hotel, weather, and budget specialists, with an input guardrail that blocks invalid or harmful requests before execution.
- Integrated **MCP** tool servers for **Tavily**, **AviationStack**, and a custom **OpenWeather** server, with isolated tool calls, normalized outputs, and fallback API-key retry handling for resilient external-data access.
- Implemented **HITL** itinerary approval with revision loops and persistent thread state using a **PostgreSQL LangGraph checkpointer**; exposed the workflow through FastAPI and packaged the application with Docker.

Relevant Coursework

Artificial Intelligence, Data Science (Machine Learning, Deep Learning, Natural Language Processing), Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming

Achievements

- **Runner-Up, HackAthena 2025** - placed among the top finalists from 1,259 registered teams at a national-level hackathon.
- Solved **300+** Data Structures and Algorithms problems in C++, covering arrays, strings, trees, graphs, recursion, and dynamic programming.